SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO4:
Design network topologies star, mesh, bus, and hybrid using networking devices, transmission media, and cabling standards
(BL2,BL6)

Assessment Tool Weightage: 30%

Annexure A

Industry Problem Solving Task

Code of Conduct:

I **C.BHUVANESH (192521108)** *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: c.bhuvanesh

Annexure A

Industry Problem Solving Task

1. A star topology is ideal for a 20-person startup because all devices connect to a central switch, making the network easy to manage and troubleshoot. At the core sits a Layer 2/Layer 3 managed switch (such as a Cisco Catalyst or Netgear ProSafe) with at least 24 ports to accommodate all workstations, printers, and VoIP phones with room to grow. Each device connects to the switch using Cat6 UTP cables, which support Gigabit speeds up to 100 meters — more than sufficient for any typical office floor plan. The switch uplinks to a router/firewall (such as a Cisco RV series or pfSense appliance) that handles internet access, DHCP, and network address translation. A Wi-Fi access point connected to the switch extends wireless coverage for laptops and mobile devices. The key advantage of this design is fault isolation: if one workstation's cable fails, only that single node goes offline, leaving all other 19 employees unaffected. Centralised cable management also simplifies future moves, adds, and changes.

A **star topology** is a good choice for a 20-person startup because every computer and network device connects to a **central switch**.

Network Structure

```
        Internet
           |
      Router/Firewall
           |
     Central Switch
   /  /  |  \  \
 PC1  PC2  PC3  Printer  VoIP

           |
       Wi-Fi AP
      /  |  \
  Laptop Phone Tablet
```

Main Components

1. **Managed Switch**
    - A 24-port Layer 2/Layer 3 switch can connect the 20 employees' devices.
    - The extra ports allow future expansion.
    - It makes network management and troubleshooting easier.
2. **Cat6 UTP Cables**
    - Each device connects to the switch using Cat6 cable.

    - Cat6 supports **Gigabit Ethernet** over typical office distances, with Ethernet channel limits commonly reaching up to 100 meters.
3. **Router/Firewall**
    - Connects the office network to the Internet.
    - Provides functions such as **DHCP, NAT, routing, and firewall protection**.
4. **Wi-Fi Access Point**
    - Provides wireless connectivity for laptops, smartphones, and tablets.
    - It connects to the central switch.

Main Advantage: Fault Isolation

The biggest advantage is that a failure in one connection normally affects **only that device**.

For example:

```
PC1 ───┐
PC2 ───┤
PC3 ───┤
PC4 ───┤── Central Switch ── Router ── Internet
PC5 ───┤
PC6 ───┘
```

If the cable connecting **PC3** fails:

PC3 → ❌

PC1 → ✓

PC2 → ✓

PC4 → ✓

PC5 → ✓

PC6 → ✓

The other users can continue working.

Advantages

- **Easy to manage**
- **Easy to troubleshoot**
- **Good performance**
- **Fault isolation**
- **Easy to expand**
- **Centralized cable management**
- Suitable for **small offices and startups**

Disadvantage

The central switch is a **single point of failure**. If the switch itself fails, all connected devices lose network connectivity. For a business that requires high availability, a backup switch or redundant design can be considered.

Conclusion

A **star topology with a managed switch, Cat6 cabling, router/firewall, and Wi-Fi access point** provides a practical, scalable, and easy-to-maintain network for a 20-person startup. It offers good performance while making troubleshooting and future expansion simple.

2. A bank demands near-zero downtime, making a full or partial mesh topology the right choice. In a full mesh, every core network device — routers, core switches, and servers — connects directly to every other device, so no single link or node failure can bring down the network. For a bank, a partial mesh is typically deployed at the core layer (connecting all core switches and firewalls to each other) while the distribution and access layers use redundant uplinks to at least two core switches. Dual enterprise-grade routers (such as Cisco ASR or Juniper MX series) provide BGP-based redundant internet connectivity. All links between core devices use fiber optic cables — either single-mode for long distances between buildings or multi-mode OM4 within the data center — because fiber is immune to electromagnetic interference, supports very high bandwidth, and is essential for maintaining low-latency financial transactions. Redundant power supplies, UPS systems, and RAID storage arrays further complement the mesh design. The justification is straightforward: when a link fails in a mesh, traffic automatically reroutes through alternative paths in milliseconds using protocols like OSPF or BGP, satisfying regulatory requirements for uptime and data integrity.

For a **bank**, network downtime can cause serious problems such as failed transactions and loss of customer trust. Therefore, a **full or partial mesh topology with redundancy** is suitable.

Network Structure

```
      Internet
      /     \
Router 1     Router 2
  | \       / |
  |  \     /  |
Core SW1 ——— Core SW2
  |\        /|
  | \      / |
Access     Access
 SW1        SW2
  |          |
Users      Servers
```

1. Partial Mesh at the Core

Instead of connecting every device to every other device, the bank can use a **partial mesh** for the important core devices.

- Core switches have redundant connections.
- Firewalls can have multiple connections.

- Critical servers can have redundant network paths.
- Access/distribution switches connect to at least two core switches.

This provides alternative paths if one connection fails.

2. Redundant Internet Routers

Use **two enterprise-grade routers** instead of one.

```
Internet
  |
+--+--+
|    |
R1   R2
|    |
+--+--+
  |
Core Network
```

If Router 1 fails, traffic can use Router 2.

Routing protocols such as **BGP** can provide redundant connectivity to external networks, while **OSPF** can be used for internal routing.

3. Fiber Optic Connections

Fiber should be used between important core devices.

- **Single-mode fiber:** Suitable for long distances between buildings.
- **Multimode fiber:** Suitable for shorter distances, such as within a data center.
- Very high bandwidth
- Low signal loss
- Resistant to electromagnetic interference
- Suitable for high-speed financial applications

4. Automatic Failover

One of the most important benefits of mesh connectivity is **alternative paths**.

For example:

Normal:

```
Core A ──────────── Core B
  \              /
   └── Core C ─────┘
```

If A → B link fails:

```
Core A ──X── Core B
  \
   └──── Core C ──── Core B
```

Traffic can use the alternate route instead of stopping completely.

5. Additional Redundancy

A bank should also eliminate other single points of failure:

- Redundant power supplies
- UPS systems
- Backup generators
- Redundant firewalls
- RAID storage
- Multiple Internet connections
- Backup network devices

Advantages

Feature	Benefit
Mesh/partial mesh	Multiple communication paths
Redundant routers	Internet connectivity continues after failure
Fiber	High speed and low interference
OSPF/BGP	Dynamic route selection
Dual core switches	Prevents core-switch failure from stopping the network
UPS/redundant power	Protects against power failures
RAID	Protects data availability

Conclusion

A **partial mesh with redundant core switches, routers, firewalls, and fiber connections** is a strong design for a bank requiring near-zero downtime. If one link or device fails, routing protocols can redirect traffic through an alternative path. This provides **high availability, fault tolerance, low latency, and reliable financial communication**.

The main disadvantage is **higher cost and complexity** compared with a simple star topology, but for a bank, the improved availability and reliability justify the investment.

3. For a small classroom lab with a tight budget, bus topology offers the lowest hardware cost because all computers share a single common communication line — the bus — eliminating the need for a central switch or hub. A single coaxial cable (RG-58, 50-ohm) or, in a modern variation, a single Cat5e/Cat6 cable daisy-chained through an inexpensive unmanaged hub runs along the length of the room. Each computer connects to the backbone using a T-connector and a BNC connector in a classic coaxial setup, with terminators (50-ohm) placed at both ends of the cable to absorb signals and prevent reflection. For a more contemporary low-cost implementation, a single inexpensive 8- or 16-port unmanaged switch replaces the terminators and uses Cat5e UTP patch cables to each workstation — achieving bus-like simplicity at minimal cost. The primary limitation is that the network is susceptible to collisions and a single cable break disrupts all connected machines, but for a supervised classroom environment with light usage (internet browsing, file sharing), this trade-off is entirely acceptable. No dedicated server is required; a teacher's PC can share resources using built-in Windows/Linux file sharing.

For a **small classroom lab with a limited budget**, a traditional **bus topology** can be considered because it requires less networking hardware than a star topology.

Basic Structure

In a traditional bus network, all computers share one main cable:

```
Terminator                    Terminator
  |                              |
  ↓                              ↓
---+--------+--------+--------+--------+------------
  |     |     |     |     |
 PC1   PC2   PC3   PC4   PC5
```

The main cable is called the **backbone**.

1. Low Cost

A traditional coaxial bus network does not require a central switch for every computer.

Typical components include:

- Coaxial cable
- T-connectors
- BNC connectors
- 50-ohm terminators
- Network adapters

This can make the setup inexpensive for a small lab.

2. Terminators

A traditional coaxial bus requires a **terminator at both ends** of the cable.

```
50Ω Terminator                    50Ω Terminator
     ↓                                 ↓
   [T]----[T]----[T]----[T]----[T]
    |      |      |      |      |
   PC1    PC2    PC3    PC4    PC5
```

The terminators prevent signal reflections from interfering with communication.

3. Advantages

- **Low hardware cost**
- Simple physical design
- Uses less cabling than a star network
- Suitable for a small number of computers
- Easy to understand for educational purposes

4. Disadvantages

The major problem is that the **main cable is a single point of failure**.

If the backbone breaks:

PC1 —— PC2 —— ❌ —— PC3 —— PC4

Communication between the affected sections can fail.

Other disadvantages include:

- Greater possibility of collisions
- Troubleshooting can be difficult
- Performance decreases as traffic increases
- Limited scalability
- Difficult to expand compared with modern switched networks

5. Modern Low-Cost Alternative

In a **real classroom today**, I would normally recommend a small **unmanaged Ethernet switch with Cat5e/Cat6 cables** rather than constructing a traditional coaxial bus.

```
      Unmanaged Switch
       / / | \ \
      / /  |  \ \
    PC1 PC2  PC3  PC4 PC5
```

This is technically a **star topology**, not a bus, but it is inexpensive, easier to maintain, and much more reliable.

Comparison

Feature	Traditional Bus	Small Switched Network
Cost	Very low	Low
Central switch	Not required	Required
Reliability	Low	Higher
Troubleshooting	Difficult	Easier
Collisions	More likely	Greatly reduced
Scalability	Poor	Better
Modern usage	Rare	Common

Conclusion

A traditional **bus topology** can be suitable for a small classroom lab when the **main priority is minimizing cost** and network usage is light. However, it is largely obsolete in modern Ethernet networks. For an actual classroom deployment, a small **unmanaged switch with Cat5e/Cat6 cabling** would generally provide a better balance of **cost, reliability, performance, and ease of maintenance**.

4. A large university campus serves thousands of users across multiple buildings — lecture halls, labs, libraries, dormitories, and administration — making a single topology inadequate. A hybrid topology combines the best elements of multiple designs: a fiber-based ring or mesh at the campus backbone level for redundancy, a star topology within each building for manageability, and bus or tree structures within individual labs for cost-efficiency. At the campus core, high-speed fiber optic cables (single-mode, 10 Gbps or 40 Gbps) interconnect core routers and multilayer switches using a ring or partial mesh arrangement, ensuring that no single link failure isolates an entire building. Each building has its own distribution switch connected to the campus core via dual fiber uplinks. Within each building, a traditional star topology branches from the distribution switch to floor-level access switches, which then connect individual devices via Cat6 cables. This hierarchical three-layer model — core, distribution, and access — is the industry standard for enterprise and campus networks because adding a new building simply means installing a new distribution switch and running fiber to the core, with no disruption to existing infrastructure. Wireless access points throughout the campus connect via PoE (Power over Ethernet) to access switches, eliminating the need for separate power cabling. The hybrid design thus delivers redundancy at the backbone, simplicity at the building level, and scalability campus-wide.

A **hybrid topology** is the best choice for a large university because thousands of users are spread across different buildings. Instead of using one topology everywhere, different topologies are combined according to the requirements of each area.

1. Overall Structure

```
                 Internet
                    |
               Core Routers
              /    |     \
             Core Switches
            /     |      \
          /       |       \
     Building A  Building B  Building C
        |          |          |
   Distribution  Distribution  Distribution
     Switch        Switch        Switch
     /  \          /  \          /  \
  Access APs     Access APs    Access APs
   Switches       Switches      Switches
    / | \          / | \         / | \
 PCs PCs PCs    PCs PCs PCs   PCs PCs PCs
```

2. Campus Core — Ring/Partial Mesh

The university's backbone should use **high-speed fiber optic connections** between core routers and multilayer switches.

A ring or partial mesh provides multiple paths:

```
     Core 1
     /   \
    /     \
 Core 2 ─── Core 3
```

If one fiber link fails, traffic can use another path. This provides **high reliability and fault tolerance**.

3. Building Distribution Layer

Each building should have a **distribution switch** connected to the campus core using **dual fiber uplinks**.

For example:

```
        Core
        /   \
    Fiber     Fiber
      /         \
```

```
     Dist Switch
         |
    Access Switches
```

If one uplink fails, the second uplink can continue carrying traffic.

4. Building-Level Star Topology

Inside each building, a **star topology** is suitable.

```
        Distribution Switch
         /   |   |   \
        /    |   |    \
      SW1    SW2   SW3   SW4
     /|\ /|\
    PC PC PC PC PC PC
```

This makes the network:

- Easy to manage
- Easy to troubleshoot
- Easy to expand
- Suitable for large numbers of devices

5. Cat6 Cabling

Access switches can connect computers, printers, and other wired devices using **Cat6 Ethernet cables**.

Cat6 provides suitable Gigabit-class connectivity for typical horizontal office/classroom cabling distances.

6. Wireless Network

Wireless access points can be connected to access switches using **PoE (Power over Ethernet)**.

```
Access Switch
   |
  PoE Cable
   |
Wi-Fi Access Point
  /  |  \
Laptop Phone Tablet
```

PoE allows the same Ethernet cable to carry **both data and electrical power**, reducing the need for separate power cables at each AP location.

7. Three-Layer Design

The network can follow the standard hierarchical model:

Layer	Main Function
Core	Fast backbone communication
Distribution	Routing, policy, and redundancy
Access	Connects users and end devices

8. Why Hybrid Topology is Suitable

Reliability:
Redundant fiber links at the core prevent one link failure from disconnecting buildings.

Manageability:
Star topology inside buildings makes troubleshooting easier.

Scalability:
A new building can be connected by adding a distribution switch and fiber uplinks.

Performance:
High-speed fiber provides high-capacity backbone connectivity.

Cost Efficiency:
Expensive redundant infrastructure is concentrated in the backbone, while simpler switching is used closer to users.

Conclusion

A **hybrid topology combining partial mesh/ring at the campus core, redundant fiber uplinks at the distribution layer, and star topology at the access layer** is ideal for a large university. It provides a good balance of **reliability, performance, manageability, cost, and scalability**. Wireless APs connected through PoE further provide flexible campus-wide connectivity.

5. Designing a network for a company with multiple departments involves creating a structured and scalable architecture using switches and routers to ensure efficient communication and security. Each department (such as HR, Finance, Sales, and IT) is connected through dedicated Layer 2 switches that form the access layer, allowing devices like computers and printers to communicate within the same department. These switches are then connected to a central Layer 3 switch or router, which acts as the core layer and enables communication between different departments through inter-VLAN routing. Virtual LANs (VLANs) are configured to logically separate each department’s network, improving security and reducing broadcast traffic. A router is also used to connect the internal network to external networks such as the internet, often with firewall functionality for security. Redundant links and backup paths can be included to ensure reliability, while proper IP addressing and subnetting are applied for efficient network management. This hierarchical design improves performance, enhances security, and allows easy expansion as the company grows.

A company with departments such as **HR, Finance, Sales, and IT** should use a **hierarchical network with VLANs**. This provides better security, performance, management, and scalability.

1. Network Structure

```
                 Internet
                    |
               Router/Firewall
                    |
             Layer 3 Core Switch
            /     |     |     \
           /      |     |      \
      HR VLAN  Finance  Sales    IT VLAN
        |       VLAN    VLAN       |
      L2 Switch  L2 Switch L2 Switch L2 Switch
       / | \     / | \    / | \   / | \
     PCs PCs    PCs PCs   PCs PCs   PCs PCs
```

2. Department VLANs

Each department is assigned a separate **VLAN**.

Department	Example VLAN	Purpose
HR	VLAN 10	HR employees
Finance	VLAN 20	Financial systems
Sales	VLAN 30	Sales employees
IT	VLAN 40	IT administrators

VLANs logically separate departments even when they use the same physical switching infrastructure.

3. Layer 2 Access Switches

Each department can have an access switch.

For example:

HR PCs ──→ HR Access Switch ──→ Core Switch

The switch connects:

- Computers
- Printers
- IP phones
- Other department devices

Devices within the same VLAN can communicate directly at Layer 2.

4. Layer 3 Core Switch

A **Layer 3 switch** performs **inter-VLAN routing**.

For example:

```
HR VLAN
  |
  +── Layer 3 Switch ── Finance VLAN
          |
          +── Sales VLAN
          |
          +── IT VLAN
```

If an HR computer needs to communicate with a Finance server, the Layer 3 switch routes the traffic between the VLANs.

5. Internet Connection

A router/firewall connects the internal network to the Internet.

```
Department VLANs
      |
  Layer 3 Switch
      |
 Router/Firewall
      |
  Internet
```

The firewall can control which internal and external traffic is permitted.

6. Security

VLANs improve security by separating departments.

For example:

- HR → HR resources: **Allow**
- Finance → Financial servers: **Allow**
- Sales → Finance network: **Restrict**
- Guest network → Internal network: **Deny**

Access-control rules can be configured on the Layer 3 switch or firewall.

7. IP Addressing and Subnetting

Each VLAN should have its own subnet.

Example:

VLAN	Department	Subnet
VLAN 10	HR	192.168.10.0/24
VLAN 20	Finance	192.168.20.0/24
VLAN 30	Sales	192.168.30.0/24
VLAN 40	IT	192.168.40.0/24

This makes network management and troubleshooting easier.

8. Redundancy

For better reliability, the company can add:

- Two core switches
- Redundant switch uplinks
- Backup Internet connection
- Redundant routers/firewalls

If one link or device fails, traffic can use the backup path.

Advantages

Performance: VLANs reduce unnecessary broadcast traffic.

Security: Departments are logically isolated.

Scalability: New departments can be added by creating additional VLANs and subnets.

Management: Centralized Layer 3 routing makes the network easier to control.

Reliability: Redundant links and devices reduce downtime.

Conclusion

A **hierarchical network using Layer 2 access switches, a Layer 3 core switch, VLANs, subnetting, and a router/firewall** is an effective design for a company with multiple departments. It provides **secure communication, efficient traffic management, easy troubleshooting, reliability, and future scalability**.

RUBRICS FOR CALCULATION

Criteria	Weightage	Exemplary (4)	Proficient (3)	Developing (2)	Beginning (1)
Problem Analysis	15%	Clearly identifies and deeply analyzes the industry problem with all factors	Good understanding with minor gaps	Basic understanding; some key aspects missing	Poor or incorrect understanding of the problem
Solution Design	20%	Innovative, practical, and industry-relevant solution	Effective solution with minor limitations	Partially suitable solution	Inappropriate or unclear solution
Technical Implementation	20%	Fully functional, accurate, and well-executed implementation	Mostly functional with minor errors	Partially implemented solution	Incomplete or non-functional implementation
Use of Tools & Technologies	10%	Appropriate and advanced use of relevant tools and technologies	Correct use of tools	Limited or basic use of tools	Incorrect or no use of tools
Problem-Solving Approach	10%	Logical, systematic, and well-structured approach	Mostly logical approach	Somewhat disorganized approach	No clear approach
Innovation & Creativity	10%	Highly creative and original solution	Some creativity	Limited originality	No creativity
Testing & Validation	5%	Thorough testing with real-world scenarios	Adequate testing	Minimal testing	No testing
Documentation & Presentation	5%	Clear, professional, and well-documented	Good documentation	Basic documentation	Poor or no documentation
Teamwork / Collaboration	5%	Excellent coordination and contribution (if team-based)	Good collaboration	Limited participation	No collaboration or contribution